Practice 3
Equation of State of a Lennard-Jones Fluid

Objective: To determine the Equation of State of a Lennard Jones fluid for a given isotherm.

Main task: To create a personal monte carlo (MC) code to simulate a Lennard-Jones (LJ) system. Use N ~ 500 atoms.

Outline

1. Write a MC code for a (N,V,T) simulation of a Lennard-Jones system . You can use the leapfroplj.f program as the starting point.

2. Compute the interaction energy (U) and the pressure (P) of the system by using the expressions.

$$U = < \sum_i \sum_{j>i} U(r_{ij}) >$$

$$P = \rho k_B T + \frac{1}{3V} < \sum_i \sum_{j>i} \vec{r}_{ij} \cdot \vec{F}_{ij} >$$

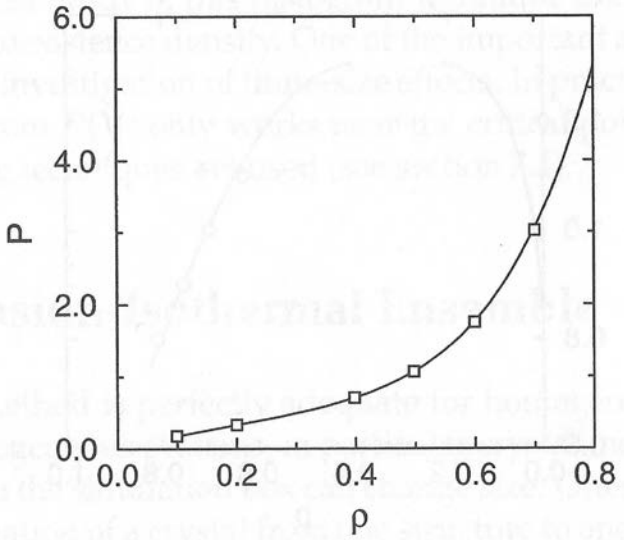
- ✓ *Calculate the standard deviation (σ_U and σ_P) of both quantities*

3. Compute the radial distribution (g(r)) function of the system.

4. Perform a series of MC simulations at T* = 2.0 and densities from ρ*=0.1 to ρ*=0.8. Take ncycl ~ 10^6, 10^7. List the technical parameters (The number of cycles ncycl, the displacement Δ and the acceptance rate, the cut-off of interaccions r_c) and the results (U, σ_U, P , σ_P).

5. Plot the quantities U* and P* as a function of ρ*. Up to what density is the ideal gas law $P = \rho k_B T$ a good approximation?

Some results from Frenkel&Smith's book:



6. Plot the functions g(r) obtained for the different densities.

7. The energy and pressure can also be obtained by numerical integration of the RDF. Make the numerical integrations, and compare the results with the direct calculation of these properties carried out in Step 4.

$$U = 2\pi\rho N \int_0^\infty U(r) g(r) r^2 dr$$

$$P = \rho k_B T - \frac{2\pi}{3}\rho^2 \int_0^\infty \frac{dU(r)}{dr} g(r) r^3 dr$$

Some comments / remarks

- Calculating the Boltzmann factor $\exp[-\beta\, U(r^N)]$ in reduced units:

$$\beta^* U^* = \beta U = \beta(\varepsilon U^*) \Rightarrow \beta^* = \beta\varepsilon$$

$$\beta^* = \beta\varepsilon = \frac{1}{k_B T}\varepsilon = \frac{1}{k_B T/\varepsilon} = \frac{1}{T^*}$$

- If a trial move is rejected, should we count the old configuration again ?

YES, because the transition probability must be normalized !!

$$\sum_n \pi(o \to n) = 1 \quad \Rightarrow \quad \pi(o \to o) = 1 - \sum_{n \neq o} \pi(o \to n)$$

- Remember the criterium to choose the displament parameter Δ: to obtain an acceptance of 50 %. Try Δ* values among 0.05 and 2.

- When computing energy and pressure, do not forget to add the tail corrections !!

$$U_{LJ}^{*,tail} = \frac{8\pi}{3} N\rho^* \left[\frac{1}{3} \left(\frac{1}{r_c^*} \right)^9 - \left(\frac{1}{r_c^*} \right)^3 \right]$$

$$P_{LJ}^{*,tail} = \frac{16}{3} \pi \rho^{*2} \left[\frac{2}{3} \left(\frac{1}{r_c^*} \right)^9 - \left(\frac{1}{r_c^*} \right)^3 \right]$$

Some results to compare with:

Table 3. Monte Carlo simulation results of this work.

$\rho*$	T^*	P^*	U^*	r_c/σ
0·5	5·0	4·654(3)	−2·365(1)	4·0
0·1	2·0	0·1777(2)	−0·667(1)	5·0
0·2	2·0	0·3290(6)	−1·306(1)	5·0
0·4	2·0	0·705(1)	−2·538(2)	5·0
0·5	2·0	1·069(3)	−3·149(2)	5·0
0·6	2·0	1·756(7)	−3·746(2)	4·71
0·7	2·0	3·024(7)	−4·300(2)	4·47
0·8	2·0	5·28(1)	−4·753(3)	4·27
0·9	2·0	9·09(2)	−5·030(4)	4·11
0·1	1·2	0·0776(1)	−0·840(2)	4·0
0·1	1·15	0·0707(1)	−0·869(1)	4·0
0·05	1·0	0·0369(1)	−0·478(1)	5·0
0·6	1·0	−0·269(9)	−4·228(6)	4·71
0·7	1·0	0·019(10)	−4·890(2)	4·47
0·8	1·0	1·03(1)	−5·533(1)	4·27
0·8	1·0	1·009(3)	−5·5357(6)	4·0
0·9	1·0	3·24(2)	−6·062(4)	4·11

In this table, U is the interaction energy per atom !!

J. K. Johnson, J. A. Zollweg, K. E. Gubbins,

"The Lennard-Jones equation of state revisited"

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